

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year 2020 – 2021 (Even Semester)

Degree, Semester & Branch: II Semester B.E. EEE

Course Code & Title: EE8251 Circuit Theory

Name of the Faculty member (s): Mr. A. Arun Kumar

Innovative Practice Description

- **UNIT I BASIC CIRCUITS ANALYSIS**

- **Topic:** Resistive element, Ohm's Law

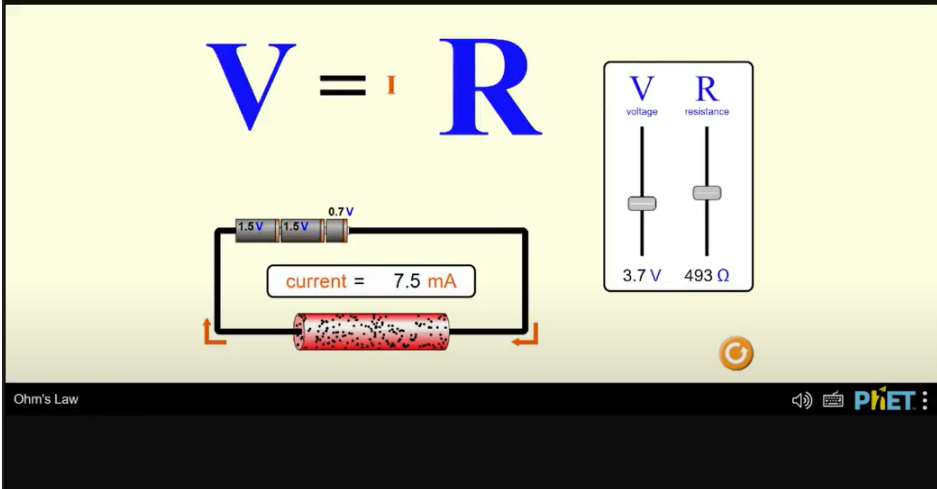
- **Activity Chosen:** Virtual Lab

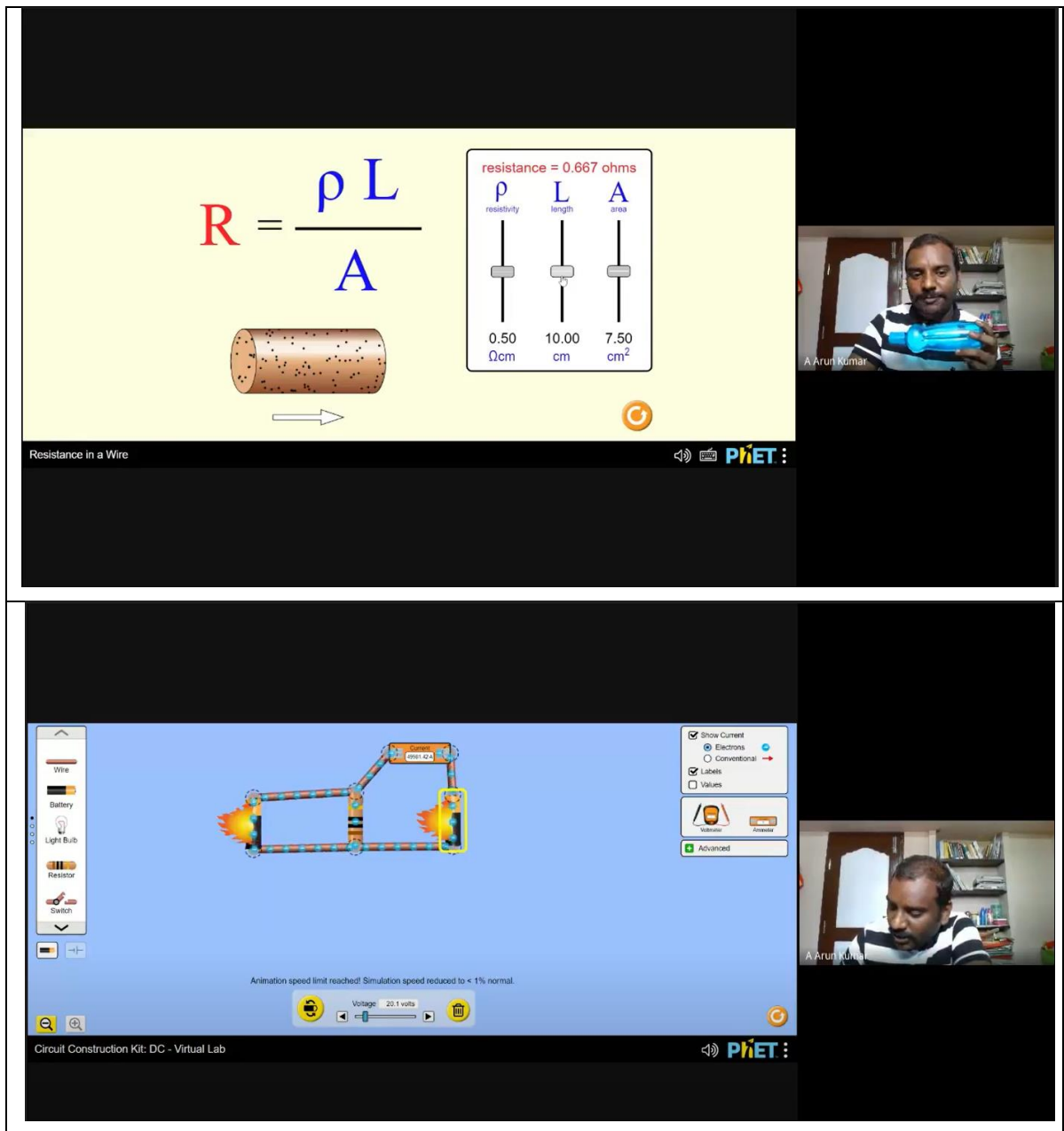
Justification for choosing this method:

Virtual lab refers to a virtual teaching and learning environment aimed at developing students' laboratory skills. As one of the most important eLearning tools, they allow the student to conduct various experiments without any constraints to place or time, in contrast to the constraints of real labs. By this students can visualize the effect of resistance in the circuit.

Time Allotted for the Activity: 20 minutes

- **Images / Screenshot of the practice:**

Innovative Teaching Method Execution	
Resistive Element, Ohm's Law – Virtual Lab	
	



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
 - ✓ Also they can able to visualise the effect of connecting the two voltage source in parallel, resistance of the resistive element.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The process of simulating a scenario to practice different responses and actions to a real life situation is extremely effective in knowledge retention. This is because knowledge isn't in theory – the user needs to apply it in a real-life situation. Simulations can be slowed down to study behavior more closely.

Conditions can be varied and outcomes investigated. Also Critical situations can be investigated without risk.

❖ ***Challenges faced in implementation:***

I planned the activity for 20 minutes. But in Class room it takes 25 minutes.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2	1	-	-	-	-	-	-	1

References:

1. William H. Hayt Jr, Jack E. Kemmerly and Steven M. Durbin, "Engineering Circuits Analysis", McGraw Hill publishers, edition, New Delhi, 2013.
2. Charles K. Alexander, Mathew N.O. Sadiku, "Fundamentals of Electric Circuits", Second Edition, McGraw Hill, 2013.
3. Allan H. Robbins, Wilhelm C. Miller, "Circuit Analysis Theory and Practice", Cengage Learning India, 2013.

CO1: After completion of the course, it is expected that students will be able to analyse the circuits using ohm's law, Kirchhoff's laws, mesh current and Node Voltage method.

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UNIT I BASIC CIRCUITS ANALYSIS

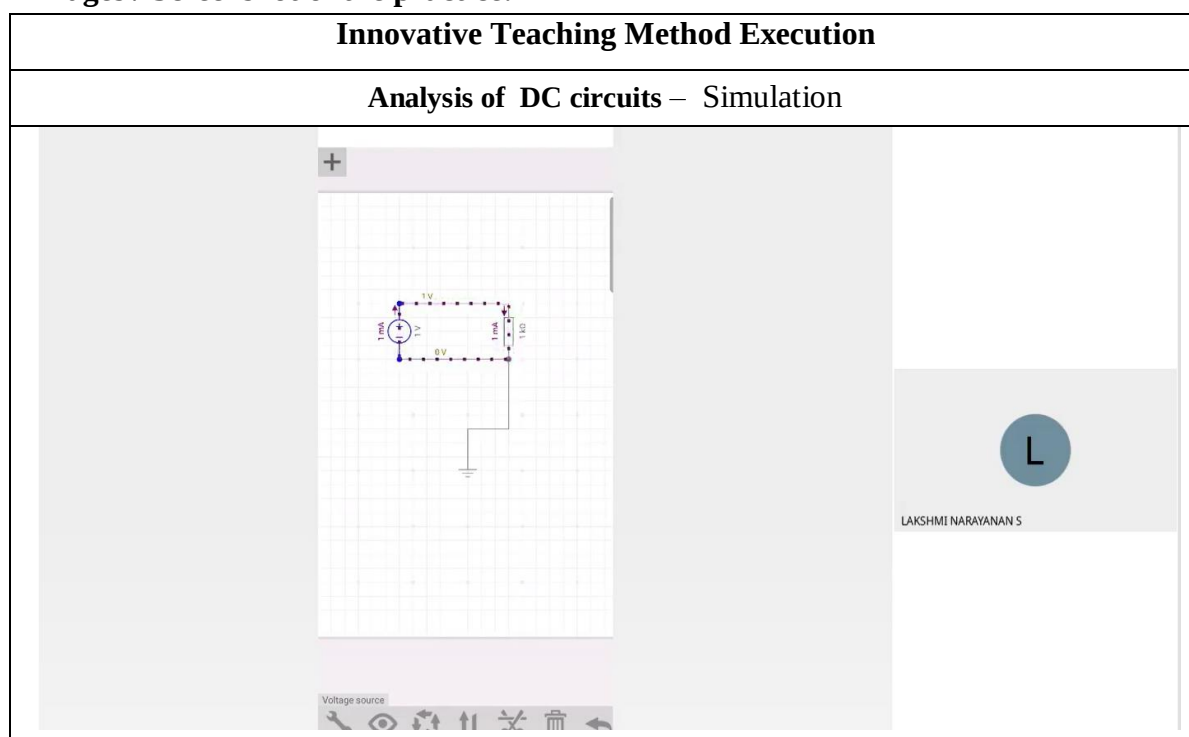
- **Topic:** Analysis of DC circuits
- **Activity Chosen:** Simulation using EveryCircuit

Justification for choosing this method:

The students will have better understanding after the usage of simulation software for the analysis of various parameters in the circuits without any risk.

Time Allotted for the Activity: 15 minutes

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- ✓ Also they can able to visualise the effect of connecting the various elements in the circuit and their parameters were measured.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

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